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CAMBRIDGE International Education

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CENTRE NUMBER

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MATHEMATICS **0580/22**

Paper 2 Non-calculator (Extended) **May/June 2026**

2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages.

DC (DE/TC) 355658/3

[Turn over

List of formulas

Area, A, of triangle, base b, height h. $A = \frac{1}{2}bh$

Area, A, of circle of radius r. $A = \pi r^2$

Circumference, C, of circle of radius r. $C = 2\pi r$

Curved surface area, A, of cylinder of radius r, height h. $A = 2\pi rh$

Curved surface area, A, of cone of radius r, sloping edge l. $A = \pi rl$

Surface area, A, of sphere of radius r. $A = 4\pi r^2$

Volume, V, of prism, cross-sectional area A, length l. $V = Al$

Volume, V, of pyramid, base area A, height h. $V = \frac{1}{3}Ah$

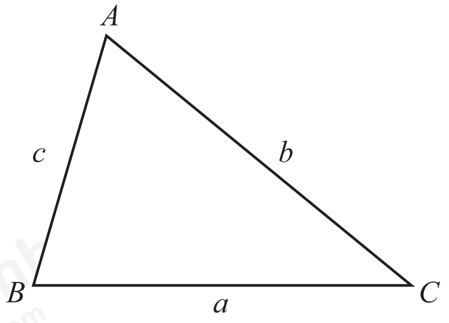
Volume, V, of cylinder of radius r, height h. $V = \pi r^2 h$

Volume, V, of cone of radius r, height h. $V = \frac{1}{3}\pi r^2 h$

Volume, V, of sphere of radius r. $V = \frac{4}{3}\pi r^3$

For the equation $ax^2 + bx + c = 0$, where $a \neq 0$, $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

For the triangle shown,



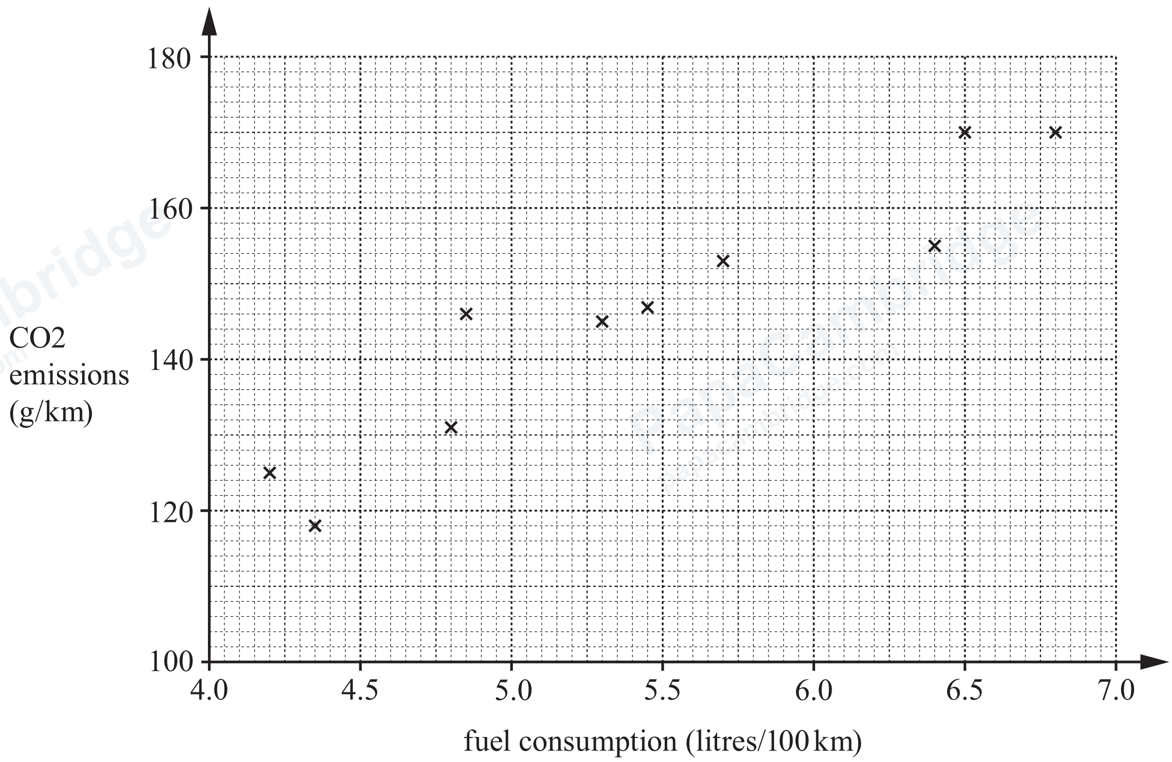
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$

Calculators must **not** be used in this paper.

1 The scatter diagram shows the fuel consumption and CO2 emissions for 10 cars.



(a) Another car has a fuel consumption of 6.2 litres/100 km and CO2 emissions of 162 g/km.

Plot a point on the scatter diagram to show this information. [1]

(b) What type of correlation does the scatter diagram show?

.... [1]

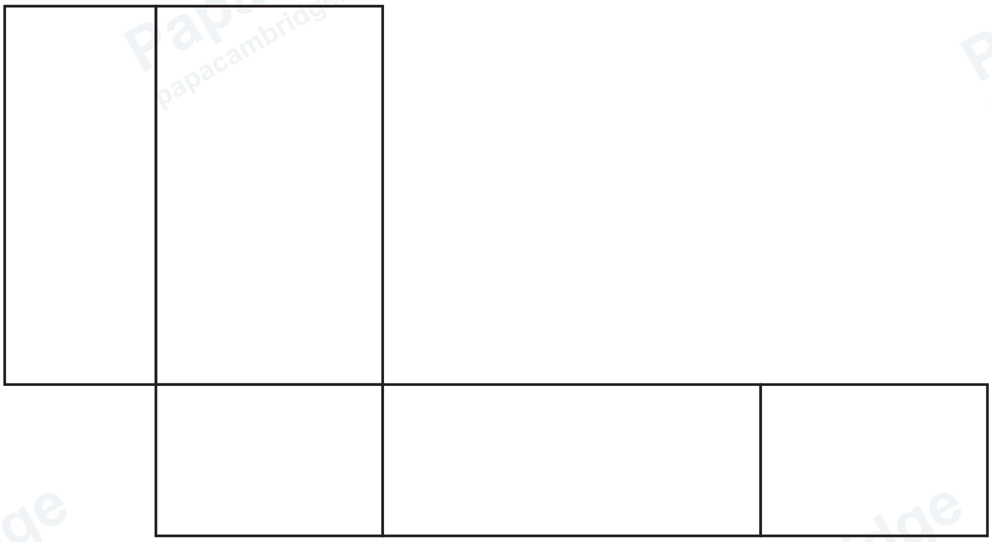
(c) Draw a line of best fit on the scatter diagram. [1]

(d) Paulo's car has CO2 emissions of 140 g/km.

Use your line of best fit to estimate the fuel consumption of his car.

.... litres/100 km [1]

2 The diagram shows an **accurate** net of an open box in the shape of a cuboid.
All the dimensions are whole numbers of centimetres.



(a) Find the total area of the net.

.... cm^2 [3]

(b) Find the volume of the open box.

.... cm^3 [2]

3 Find the reciprocal of $3\frac{1}{3}$.

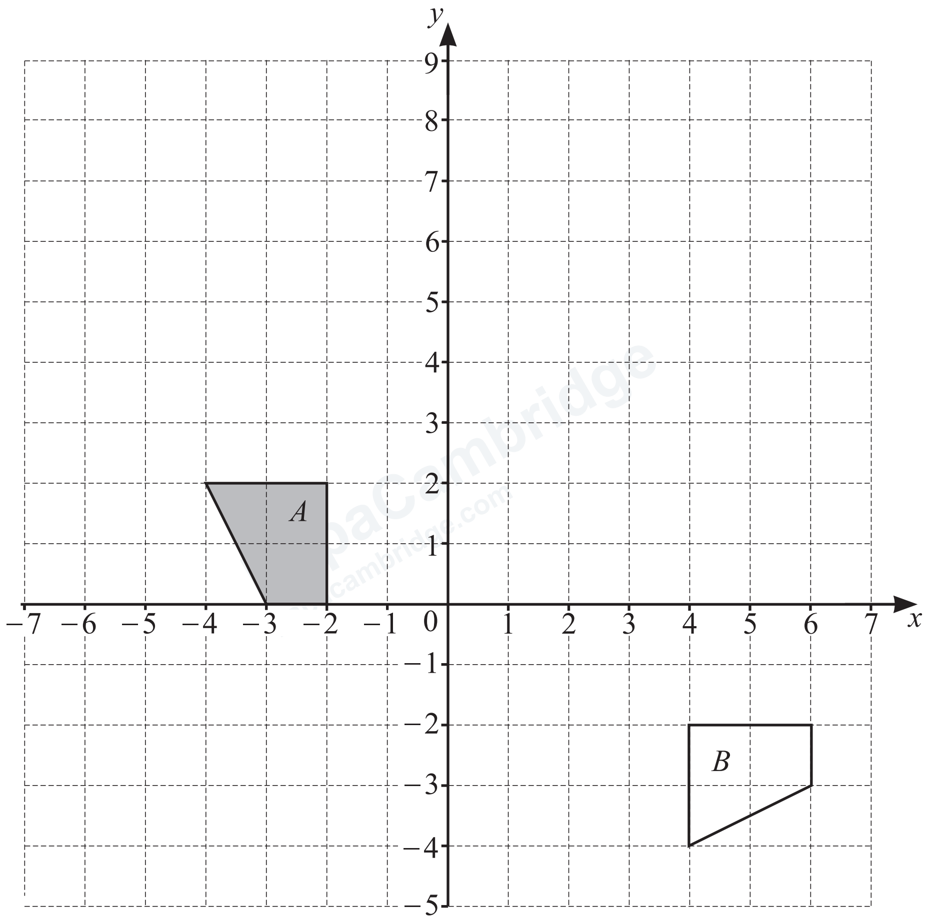
Give your answer as a fraction in its simplest form.

.... [2]

4 Work out $\frac{1}{6} \div \frac{2}{7}$.

.... [2]

5



(a) On the grid, draw the image of

(i) shape A after a translation by the vector $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$. [2]

(ii) shape A after a reflection in the line $y = x$. [2]

(b) Describe fully the **single** transformation that maps shape A onto shape B.

....

.... [3]

6 $E = 5c^2 - d^2$

(a) Find the value of E when $c = \sqrt{18}$ and $d = 2$.

$E =$ .... [2]

(b) Rearrange the formula to write c in terms of E and d.

$c =$ .... [3]

7 Factorise.

$6wx - 4xy$

.... [2]

8

$\frac{1}{\sqrt{3}}$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\sqrt{3}$	$\frac{\sqrt{3}}{2}$

Complete these statements using values from the list.

$\cos 45° =$ ....

$\tan 60° =$ .... [2]

9 **(a)** Write 63 as a product of its prime factors.

.... [2]

(b) Find the lowest common multiple (LCM) of 63 and 42.

.... [2]

(c) Find the highest common factor (HCF) of $63a^5c^2$ and $42a^2c^5$.

.... [2]

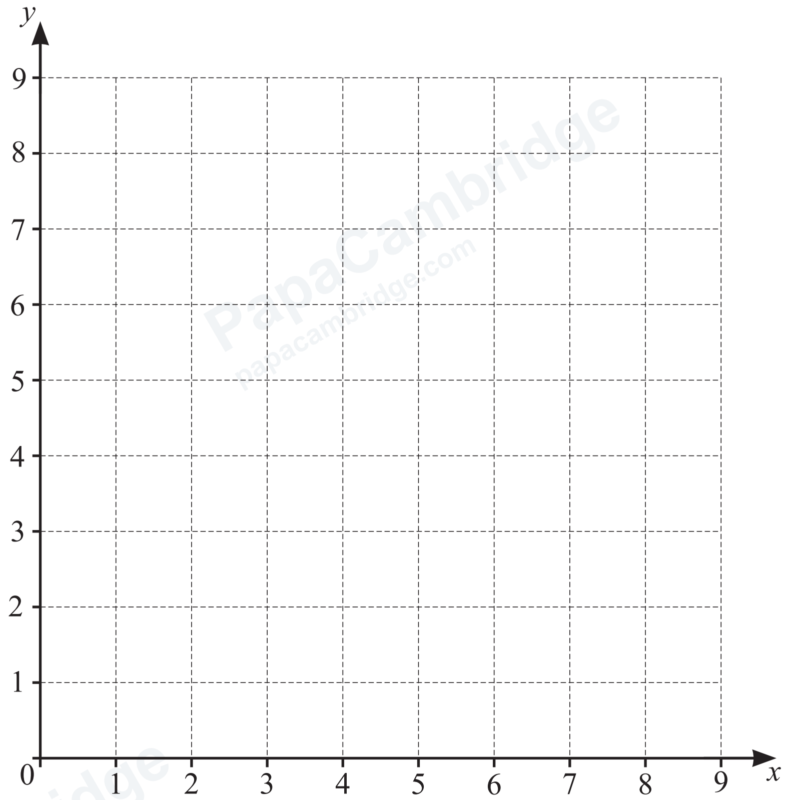
10 Write $0.7\dot{3}$ as a fraction, giving your answer in its simplest form.

.... [3]

11 Find the number of sides of a regular polygon with interior angle 140°.

.... [2]

12



(a) On the grid, draw these four straight lines.

$x = 8$
$y = 2$
$x + y = 8$
$2y - x = 4$ [5]

(b) Region R satisfies these four inequalities.

$x \leqslant 8$
$y \geqslant 2$
$x + y \geqslant 8$
$2y - x \leqslant 4$

By shading the unwanted regions of the grid, find and label the region R. [2]

DO NOT WRITE IN THIS MARGIN

13 A sector of a circle has sector angle 90° and radius 12 cm.

Calculate the arc length of the sector.
Give your answer in terms of π.

.... cm [2]

14 **(a)** The nth term of a sequence is given by $n^2 - 3$.

Write down the first three terms of this sequence.

.... [2]

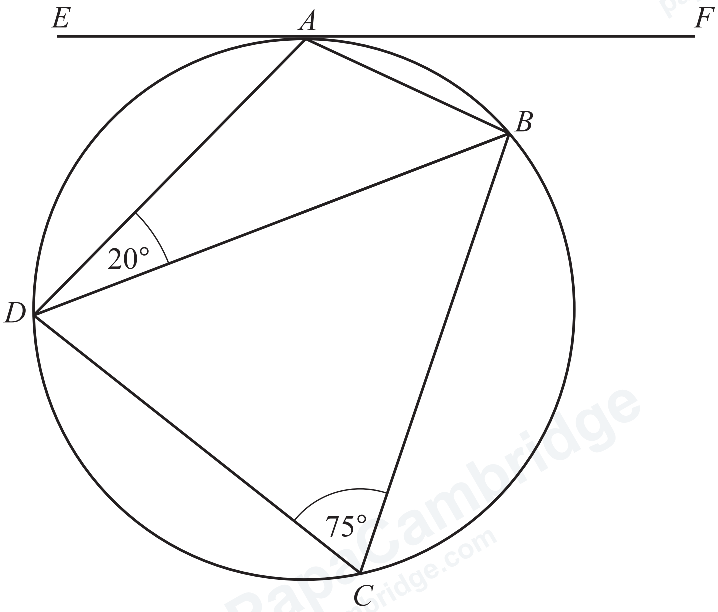
(b) These are the first four terms of another sequence.

4 12 36 108

Find the nth term of this sequence.

.... [2]

15



NOT TO SCALE

A, B, C and D are points on a circle.
EF is a tangent to the circle at A.
Angle $ADB = 20°$ and angle $BCD = 75°$.

Find angle EAD.
Give a reason for each step of your working.

angle EAD = ... [5]

16 The volume of a sphere is $500\,\text{cm}^3$.

Using the approximation of 3 for π, find an estimate for the radius of the sphere.

.... cm [2]

17 **(a)** Expand and simplify.

$$(4-\sqrt{7})(3+2\sqrt{7})$$

.... [2]

(b) Rationalise the denominator and simplify.

$$\frac{2}{3-\sqrt{5}}$$

.... [3]

18 Marwa has these 10 numbered tiles.

1 2 2 3 3 4 4 4 4 4

(a) Marwa picks one of the tiles at random.
She replaces the tile and then picks a second tile at random.

Find the probability that the two tiles have the same **even** number.

.... [3]

(b) From the 10 numbered tiles, Marwa now picks three tiles at random without replacement.

Find the probability that the sum of the numbers on the three tiles is 11.

.... [3]

19 Expand and simplify.

$$(3x+1)(x+4)(x-2)$$

.... [3]

20 Two bottles are mathematically similar.
The volume of the larger bottle is $216\,\text{cm}^3$ and the volume of the smaller bottle is $27\,\text{cm}^3$.
The height of the larger bottle is 10 cm and the height of the smaller bottle is h cm.

Find the value of h.

$h =$ .... [3]

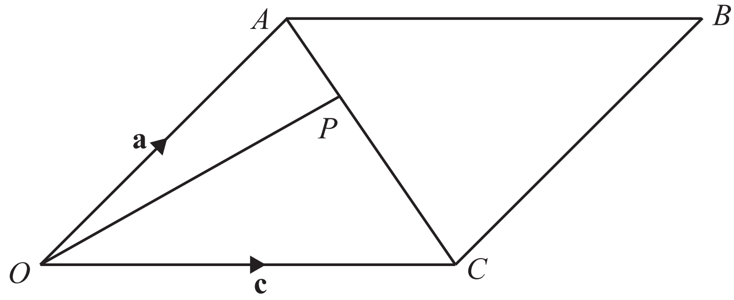
21 **(a)** Write x^2+8x-7 in the form $(x+a)^2+b$ where a and b are integers.

.... [2]

(b) Write down the coordinates of the turning point of the curve $y = x^2+8x-7$.

(.................... , ...) [1]

22



NOT TO SCALE

$OABC$ is a parallelogram.
$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$.
P is a point on AC and $AP : PC = 1 : 4$.

(a) Find, in terms of **a** and **c**, in its simplest form

(i) $\overrightarrow{AP}$

$\overrightarrow{AP} =$ .... [2]

(ii) $\overrightarrow{OP}$.

$\overrightarrow{OP} =$ .... [1]

(b) Q is a point on AB and $\overrightarrow{AQ} = \frac{1}{4}\mathbf{c}$.

Show that O, P and Q lie in a straight line.

[2]

23 The equation of a curve is $y = 2x^2 - 11x + 12$.

(a) Find the gradient of the tangent to the curve at $x = 2$.

.... [3]

(b) Line L intersects the curve at $x = 2$.
Line L is perpendicular to the tangent to the curve at $x = 2$.

Find the equation of line L.

.... [4]

24 $7^{2x+1} = 49^{2-x}$

Find the value of x.

$x =$ .... [3]

Question 25 is on the back page.

25 The straight line $y = 7 - 4x$ intersects the curve $y = 4 + 3x - 2x^2$ at two points.

Find the coordinates of these two points.

(.................... , ...)

(.................... , ...)
[5]